

Seeing Crypto Markets with Public LLMs: An Observation Layer that Compiles World Bundles

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Abstract

Purpose. Public LLMs do not need another trading strategy first; they need to *see* the market. We build a crypto *observation layer*: an anonymized world-model runtime that compiles asynchronous multi-band evidence into a point-in-time (PIT) *world bundle*—complete, honest, and auditable—for GPT /DeepSeek /GLM /Gemini-class models. Hard refusal is the safety valve of that layer, not its product definition: when support is thin, a machine-readable flag forces abstention so half-baked context cannot drive decisions. **Evidence.** On identical OOS days, a Compiled bundle makes the market legible: grounding recovers ready/missing/tilt at ≈ 0.97 vs. ≤ 0.23 under a Raw momentum fragment. The same layer’s quality gate is enforceable across four live vendors (temperature 0): Compiled thin-abstain 1.0; Ungated soft judgment on the same content only ≈ 0.68 ; Raw over-trades and loses; Blind refuses with no feed. This panel is scarce (max WMI $\approx 0.093 < 0.2$), so Compiled’s full refusal is a stress test of the valve, not a claim the layer never opens. A no-LLM content rule on vintaged tilts beats momentum ($\Delta CE = 0.334$, $p = 0.034$), showing the bundle fields are nonempty. **Claim.** Compile so models can see; refuse when they must not act.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

observation layer, world bundle, public LLMs, cryptocurrency, point-in-time, grounding, quality-gated refusal, systems

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1 Introduction

Purpose (read this first). Make public LLMs *see* crypto markets. Venue fragmentation, derivatives, macro vintages, and alternative flows arrive on incompatible clocks [11, 13]. Generative world models [6, 7, 9] and finance agents [19–21] typically *assume* a usable observation stream. We build the missing layer: compile asynchronous evidence into a consumable PIT world bundle; when the view is too thin, force abstention so half-baked context cannot harm decisions. Refusal is the safety valve—not a substitute for seeing.

What we build. An anonymized *world-model runtime* (state compiler + quality gate, not a Dreamer-style simulator) emits a JSON

world bundle for public LLMs. The bundle must be **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence IDs). Quality indices WMI/ACWMI and `should_ai_abstain` make thin worlds machine-readable.

How we evaluate. **RQ1 (primary—seeing):** Does the Compiled bundle make ready /missing /tilt answers verifiable against PIT ground truth? **RQ2 (safety valve):** When the world is thin, does a hard quality flag enforce abstention where soft disclosure does not? Four arms on identical days: Compiled (full bundle + hard flag), Ungated (same content, no flag), Raw (mom5 only), Blind (no feed). **RQ3 (secondary):** Are vintaged tilts economically nonempty under a transparent no-LLM rule?

Contributions. (1) Semantics for consumable world bundles (see before act). (2) Anonymized PIT runtime: BandPIT clock, quality-tagged bundles, four-arm harness. (3) Live validation: grounding ≈ 0.97 vs. Raw collapse; hard-gate thin-abstain 1.0 vs. Ungated ≈ 0.68 ; nonempty tilts under a no-LLM probe. No LLM trading SOTA claim.

2 Related Work

World models. World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem—a PIT-safe crypto world *before* dynamics or policy—not a generative simulator.

LLM agents in finance. Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and recent ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve skill and orchestration while assuming a usable observation stream. We study that prior layer: compile a world bundle, measure whether models can see it, then gate action when they must not act.

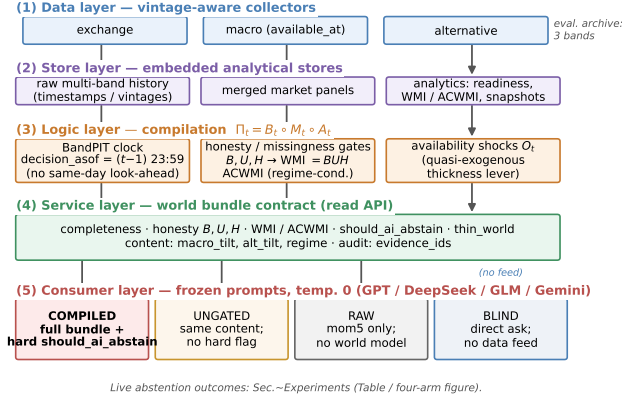
Selective prediction and PIT. Abstention can be optimal under noise [4, 5]; we attach refusal to world quality, not classifier confidence alone. Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline follows [8, 14, 17].

3 Observation Layer

Definition 3.1 (Epistemic observation). An epistemic observation is $O_{j,t} = (x, \tau, q, g, r)$: value, latest-available time, quality, main-view gate, semantic role. Omitting (τ, q, g, r) yields a feature dump, not a world observation.

Definition 3.2 (World bundle). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (WMI/ACWMI). Π_t aligns clocks, applies honesty/missingness gates, and assembles band views into a consumer JSON object. $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

Seeing vs. gating. The bundle is what the model *sees* (completeness, tilts, evidence IDs). The hard flag `should_ai_abstain` is how



Live abstention outcomes: Sec.~Experiments (Table / four-arm figure).

Figure 1: Observation-layer runtime: raw multi-band evidence → PIT world bundle → Compiled / Ungated / Raw / Blind consumers.

Table 1: Module map.

Module	Role
Collectors / stores	Vintage multi-band history
BandPIT compiler	Previous-close Π_t , WMI/ACWMI
Bundle builder	Complete / honest / auditable JSON
LLM adapter	OpenAI-compatible chat; temp. 0
Eval harness	Four-arm sampling and tables

the layer *blocks* action when $q(W_t)$ is too low. Exporting tilts without gates is still a feature dump; exporting gates without a readable bundle is only a switch.

PROPOSITION 3.3 (COMPILATION $\neq$ DUMP). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{AI}$: ungated stale evidence can raise volume while lowering honesty.*

PROPOSITION 3.4 (WORLD-CONDITIONAL ABSTENTION). *If non-abstain actions exceed a world-dependent cost, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

REMARK 3.5 (NOT A GENERATIVE WM). *We do not learn $p(s_{t+1} | s_t, a_t)$. We deliver a state compiler with a quality-gated refuse interface.*

4 Runtime

Figure 1 and Table 1 summarize the anonymized prototype. Collectors write vintage-aware exchange, macro, and alternative history; BandPIT reconstructs readiness on a previous-close clock; a service layer serves quality-tagged bundles to frozen-prompt consumers.

Evaluation archive. Three public bands: exchange (klines, funding, OI), macro (vintaged risk indices), alternative (stablecoin circulation). For day t , information is reconstructed at $(t-1)$ 23:59; payoff is same-day close-to-close return. Figure 2 shows PIT readiness: macro and alternative stay ready; exchange drops later and defines the within-archive thick/thin contrast. Listing 1 shows a Compiled

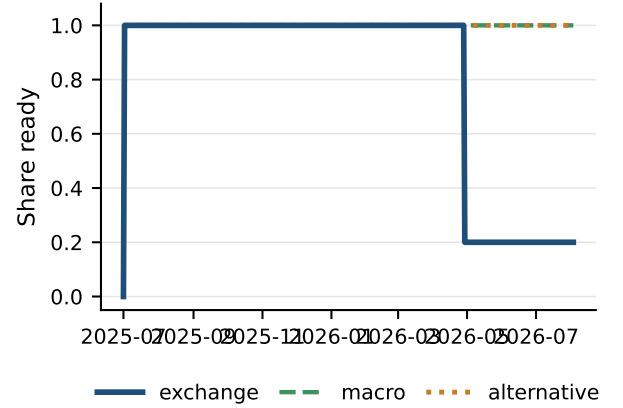


Figure 2: PIT readiness for the three-band archive (exchange readiness drops later).

bundle on an exchange-gap day: two bands ready, low WMI, hard abstain true, evidence IDs binding the view.

Listing 1: Compiled world bundle (compact OOS example).

```
{
  "date": "2026-05-11",
  "asset": "ADA",
  "decision_asof": "2026-05-10T23:59:00",
  "completeness": {
    "n_ready": 2,
    "n_missing": 1
  },
  "world_model_index": {
    "wmi": 0.015,
    "should_ai_abstain": true,
    "thin_world": true
  },
  "macro_tilt": 1.0,
  "alt_tilt": -1.0,
  "audit": {
    "evidence_ids": [
      "band:exchange:missing",
      "band:macro:ready"
    ]
  }
}
```

Four arms. Compiled: full bundle; must abstain when the hard flag is true (the frozen prompt also forbids acting when the world is judged thin). **Ungated:** same content and numeric WMI; soft “you may abstain.” **Raw:** mom5 only—common thin wiring, not a strongest-agent baseline. **Blind:** no feed. Refuse when WMI < 0.2. On this scarce panel max WMI ≈ 0.093 , every OOS day is thin under the production gate—a stress test of the valve, not proof the layer never opens.

Prompt difference (only gating). Compiled requires abstain when the hard flag is true or the world is thin; Ungated asks the model to judge thinness with no forced policy. Both see the same bundle content. That is intentional: the valve must be enforceable, not hoped for.

5 Experiments

5.1 Setup

PIT panel ~ 399 days, 10 liquid names, IS/OOS 200/200 days. Live arms share the same stratified 100 OOS asset-days (seed fixed) unless noted; temperature 0; four flash/mini vendors in Table 2. The no-LLM probe uses 199 OOS days after return alignment.

5.2 RQ1 (primary): Seeing—grounding workflow

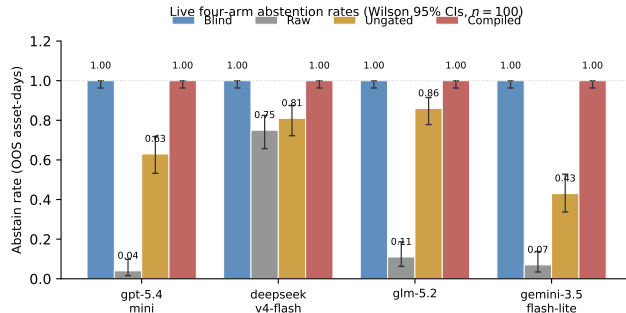
If the observation layer works, models should report what the bundle discloses. On 50 OOS asset-days, given Compiled or Raw inputs, each model must list ready bands, missing bands, and sign(macro_tilt),

Table 2: Grounding (50 days): ready / missing / tilt signs. Primary seeing result.

Model	Ready F1		Missing F1		Tilt-sign	
	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 3: Thin-world abstention on identical 100 OOS days (Wilson 95% CIs). Blind abstains 1.0. Δ CE = avoided loss.

Model	Thin C	Thin U [CI]	Abs. R [CI]	Δ CE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

**Figure 3: Safety-valve ranking: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.**

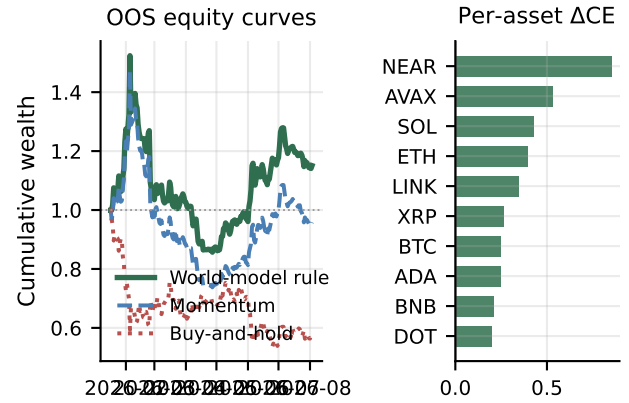
sign(alt_tilt), scored against PIT truth. Table 2: Compiled mean ready F1 0.97, missing F1 0.97, tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Sufficiency chat alone is weak—models may call a 3/3-ready day “enough” while WMI still marks the world thin. The sharp claim is *verifiability*: only the compiled bundle lets the model state a checkable view of the market.

5.3 RQ2 (safety valve): Hard refuse when the view is thin

Seeing is not enough if thin worlds still drive action. Table 3 and Figure 3: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (abstain as low as 0.04; Table 4); Blind abstains 1.0 with no feed. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean Δ CE (Compiled–Raw) +1.07 is avoided loss under the valve, not an alpha claim. Full-OOS ($n=2000$, all thin) reproduces Compiled 1.0 and Ungated mean ≈ 0.56 .

Table 4: Raw action mix (100 days): half-baked context drives directional mass.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

**Figure 4: Secondary content probe (no LLM): compiled tilts beat momentum; Δ CE positive for every asset.**

Scarce-panel note (once). Production Compiled never opens at WMI < 0.2 on this archive. On band-thick / open-slice days (3/3 ready $\equiv$ WMI ≥ 0.05 ; $n=1224$), Ungated keeps content without the hard boolean: four-vendor mean CE $\approx +0.29$; block-bootstrap mean $\approx +0.31$ with CIs including zero—a content lower bound, *not* Compiled-open trading (the prompt’s thinness clause can still force abstain). Denser vintages can raise the WMI floor without changing layer semantics.

5.4 RQ3 (secondary): Nonempty bundle fields

A transparent no-LLM rule adds vintaged macro_tilt / alt_tilt to return inputs shared with momentum. OOS: Sharpe 0.764 / CE +0.130 vs. momentum (0.096, -0.206) and buy-and-hold (-1.404 , -1.163); Δ CE=0.334 ($p=0.034$; Figure 4). LOBO attributes the gap to band content; a long non-vintaged audit shows no edge. This supports nonempty fields in the observation layer, not an LLM leaderboard.

5.5 Discussion

What carries the paper. (i) *The bundle is legible* (RQ1). (ii) *Thin views must not drive action*—soft text fails; the hard valve works (RQ2). (iii) *Fields are nonempty* under a no-LLM probe (RQ3). Relative to FinMem / FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we contribute the observation layer those systems assume.

Threats. Compiled thin abstention is quality-gate obedience by design. The archive is scarce; open-slice Ungated is not Compiled-open; CE bootstrap CIs include zero. Vendors are flash/mini at

temperature 0; Raw is common mom5 wiring for the four-arm ladder, not dominance over tool agents. Decision/execution stacks can sit on top once denser bands lift WMI above the scarce floor.

6 Conclusion

Public LLMs need a market they can see before they decide. We contribute an observation layer that compiles asynchronous crypto evidence into a consumable PIT world bundle, and forces abstention when the view is too thin. Live, models recover ready/missing/tilt at ≈ 0.97 under Compiled vs. Raw collapse; the hard valve abstains 1.0 on thin days where soft judgment does not. Compile so models can see; refuse when they must not act.

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